SIVASHAKTHI K

AI & Software Engineering | LLMs • Machine Learning • Automation • Agentic AI

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PROFESSIONAL SUMMARY

AI & Data Science graduate with hands-on experience building machine learning, NLP, and software applications using Python, TensorFlow, PyTorch, Keras, Scikit-learn, SQL, and web technologies. Experienced in developing end-to-end ML pipelines, data preprocessing, feature engineering, NLP-based conversational systems, LLM prompt engineering, automation, and database-integrated applications. Strong interest in building practical AI-powered tools, internal software, data-driven workflows, and solutions for real-world business problems.

TECHNICAL SKILLS

Programming: Python, Java, SQL

AI / ML: TensorFlow, PyTorch, Keras, Scikit-learn, Machine Learning, Deep Learning, Feature Engineering, Model Evaluation, Hyperparameter Tuning

Generative AI / NLP: LLM Prompt Engineering, Intent Classification, Entity Recognition, NLP, Sequence Models

Data: Pandas, NumPy, Data Preprocessing, Class Imbalance Handling, Data Analysis | Databases: MySQL, Database Design, JDBC

Software / Web: HTML, CSS, JavaScript, Git, Socket Programming | Other: Kali Linux, Penetration Testing

PROJECTS

Online & UPI Payment Fraud Detection

Python, Scikit-learn, TensorFlow, Pandas, MySQL

- Built an end-to-end machine learning pipeline to identify fraudulent UPI and online transactions using Random Forest and XGBoost.
- Performed data preprocessing and feature engineering to address class imbalance and improve model precision and recall.
- Tuned model hyperparameters and evaluated performance using benchmark fraud datasets and F1-score.
- Integrated MySQL for transaction logging and persistence of model outputs, supporting audit and analysis workflows.

College FAQ Chatbot

Python, NLP, Deep Learning, Keras

- Designed and deployed an NLP-based conversational assistant for student queries related to admissions, academics, departments, and placements.
- Implemented intent classification and entity recognition using deep learning sequence models.
- Improved query resolution through iterative model retraining and response refinement.

AI-Enhanced Pharmacy Medi-Track App

React Native, Expo, Firebase, Flask, Python, Machine Learning

- Built a unified AI-powered healthcare app integrating doctor appointment booking, live video consultation, and smart e-pharmacy using React Native and Firebase.
- Developed an AI-based doctor recommendation engine (Logistic Regression, Random Forest) from patient symptoms and history, with WebRTC-based encrypted video consultations.
- Built a smart pharmacy module with prescription verification and medicine reminders; published in IJSART (March 2026) and presented at ICCBIET'25.

INTERNSHIP EXPERIENCE

AI & Prompt Engineering Intern — VaultofCodes *4 Weeks*

- Designed and evaluated prompts for LLM-based applications and studied generative AI model behavior.
- Explored prompt optimization techniques to improve the quality and effectiveness of AI-generated responses.

Python Development Intern — Optimus, Salem *20 Days*

- Used Python for data manipulation, scripting, and automation tasks.
- Applied programming concepts to practical development and data-processing workflows.

Front-End Development Intern — Heal Bharat *4 Weeks*

- Built responsive web interfaces using HTML, CSS, and JavaScript.
- Applied front-end development practices to create functional and user-oriented web interfaces.

EDUCATION

B.Tech — Artificial Intelligence & Data Science *2022 – 2026*

Muthayammal Engineering College, Anna University | CGPA: 7.4

Diploma in Engineering *2023*

Muthayammal Polytechnic Institution, DOTE | 91.6%

CERTIFICATIONS & ACHIEVEMENTS

- 400+ problems solved on LeetCode | Distributed Systems — NPTEL | Micro-Certification — ServiceNow University
- IR 4.0 Technologies — Edunet Foundation | DataPlus Course Completion — TCS Mastercraft | Salesforce Trailhead: 30,000+ points, 50+ badges — Admin, Apex, LWC